

ISEN/DAEN 427

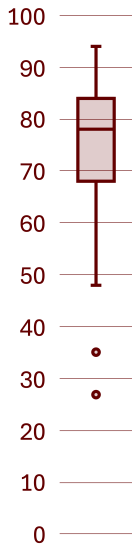
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Midterm, Credible Interval, and M&Ms

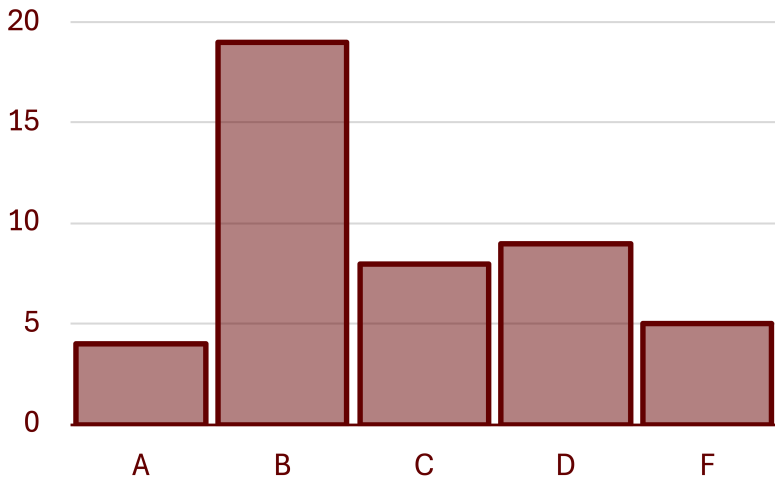
Oct 7, 2025



Midterm 1



Count of Unposted Final Grade (as Oct 7)



Beta Distribution

- The **Beta distribution** is commonly used as a prior or posterior distribution for probabilities p in Bayesian statistics.
- It is defined on the interval $[0, 1]$ with two shape parameters α and β .
- Probability density function:

$$f(p \mid \alpha, \beta) = \frac{p^{\alpha-1}(1-p)^{\beta-1}}{B(\alpha, \beta)}$$

where $B(\alpha, \beta)$ is the Beta function.

- Often used as the posterior when observing x successes out of n trials with a Binomial likelihood.

Mean and Standard Deviation

Consider a Beta distribution with parameters $\alpha > 0$ and $\beta > 0$:

$$X \sim \text{Beta}(\alpha, \beta)$$

Mean

$$\mathbb{E}[X] = \frac{\alpha}{\alpha + \beta}$$

Standard Deviation

$$\text{SD}(X) = \sqrt{\frac{\alpha\beta}{(\alpha + \beta)^2(\alpha + \beta + 1)}}$$

Bayesian Updating with Beta Distribution

- Suppose we are estimating the probability of success p in Bernoulli trials.
- We choose a **Beta prior**:

$$p \sim \text{Beta}(\alpha, \beta)$$

- We observe data: x successes out of n trials

$$\text{Likelihood: } x \sim \text{Binomial}(n, p)$$

Posterior Distribution

Using Bayes' theorem, the posterior is also Beta:

$$p \mid x \sim \text{Beta}(\alpha + x, \beta + n - x)$$

- This shows how prior beliefs (α, β) are **updated with data** to produce the posterior.
- The posterior can be used to compute credible intervals and predictive probabilities.

Credible Interval

- A **credible interval** is a Bayesian analogue of a confidence interval.
- It represents the range of parameter values that contain a specified probability mass of the posterior distribution.
- For the Beta posterior $\text{Beta}(\alpha, \beta)$, a $100(1 - \gamma)\%$ credible interval is:

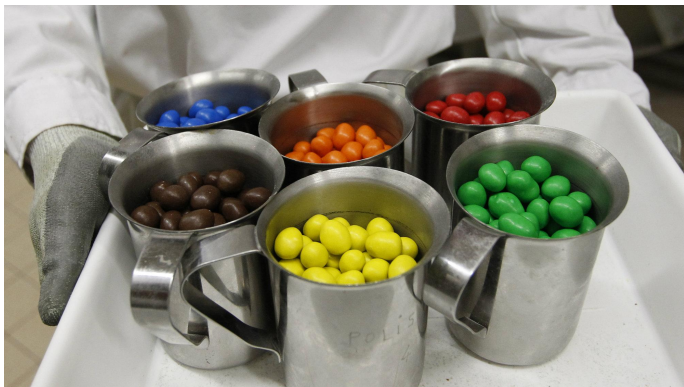
$$[q_{\gamma/2}, q_{1-\gamma/2}],$$

where q_p is the p -th quantile of the Beta distribution.

- For example, a 95% credible interval uses $\gamma = 0.05$.

Interpretation: There is a 95% probability that the true parameter p lies within the interval given the observed data.





Quiz

- What's your favorite color of M&M?
- Do you think the colors in a bag of M&Ms are random?
- Guess the proportion of your favorite color in a bag.